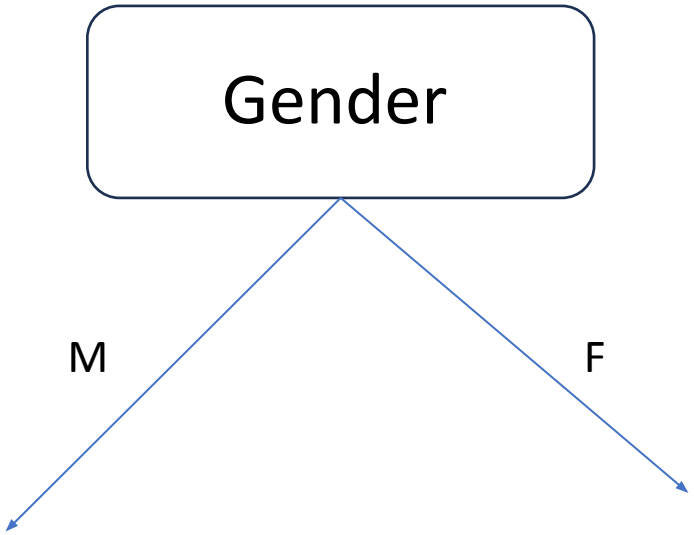
Step by Step Procedure for Regression Tree Construction

Dosage	Gender	Drug Effectiveness
4	M	0
6	M	2
10	F	3
15	F	100
20	M	100
25	M	100
30	F	20
35	F	10
40	M	0



4,6,20,25,40

10,15,30,35

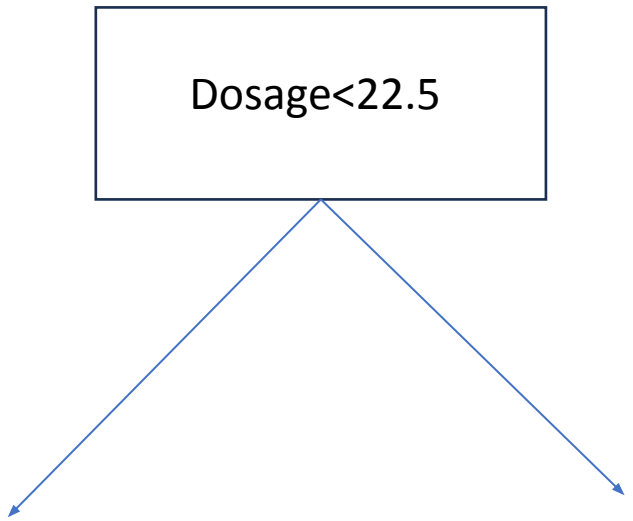
Predicted D.E = (0+2+100+100+0)/5
= 40.4

Predicted D.E = (3+100+20+10)/4
= 33.25

SSR (Gender) = (0-40.4)^2+(2-40.4)^2+(100-40.4)^2+(100-40.4)^2+(0-40.4)^2
+(3-33.25)^2+(100-33.25)^2+(20-33.25)^2+(10-33.25)^2
= 17929.95

Dosage	Gender	Drug Effectiveness
4	M	0
6	M	2
10	F	3
15	F	100
20	M	100
25	M	100
30	F	20
35	F	10
40	M	0

Threshold
5
8
12.5
17.5
22.5
27.5
32.5
37.5



4,6,10,15,20

Predicted D.E = 41

25,30,35,40

Predicted D.E = 32.5

SSR (Dosage<22.5) = (0-41)^2+(2-41)^2+(3-41)^2+(100-41)^2+(100-41)^2
+(100-32.5)^2+(20-32.5)^2+(10-32.5)^2+(0-32.5)^2
= 17883

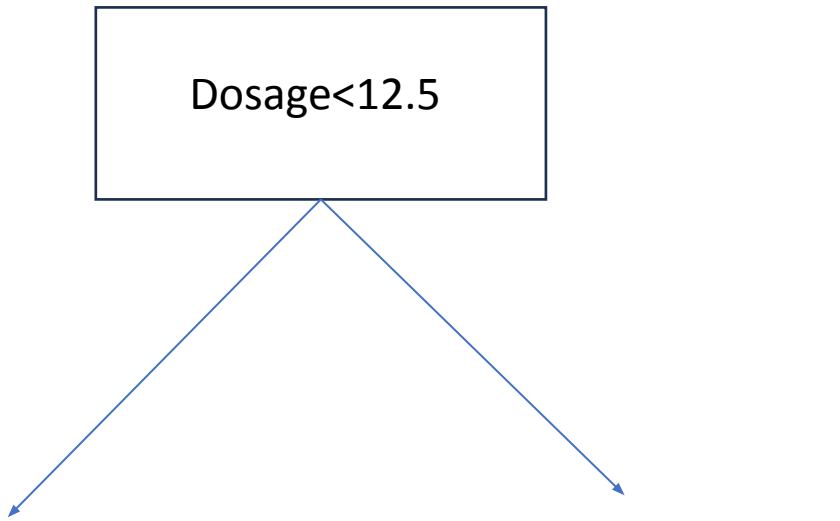
Dosage	Gender	Drug Effectiveness
4	M	0
6	M	2
10	F	3
15	F	100
20	M	100
25	M	100
30	F	20
35	F	10
40	M	0

Threshold
5
8
12.5
17.5
22.5
27.5
32.5
37.5

SSR
16485
14670
12355
17177
17883
14709
15374
16485

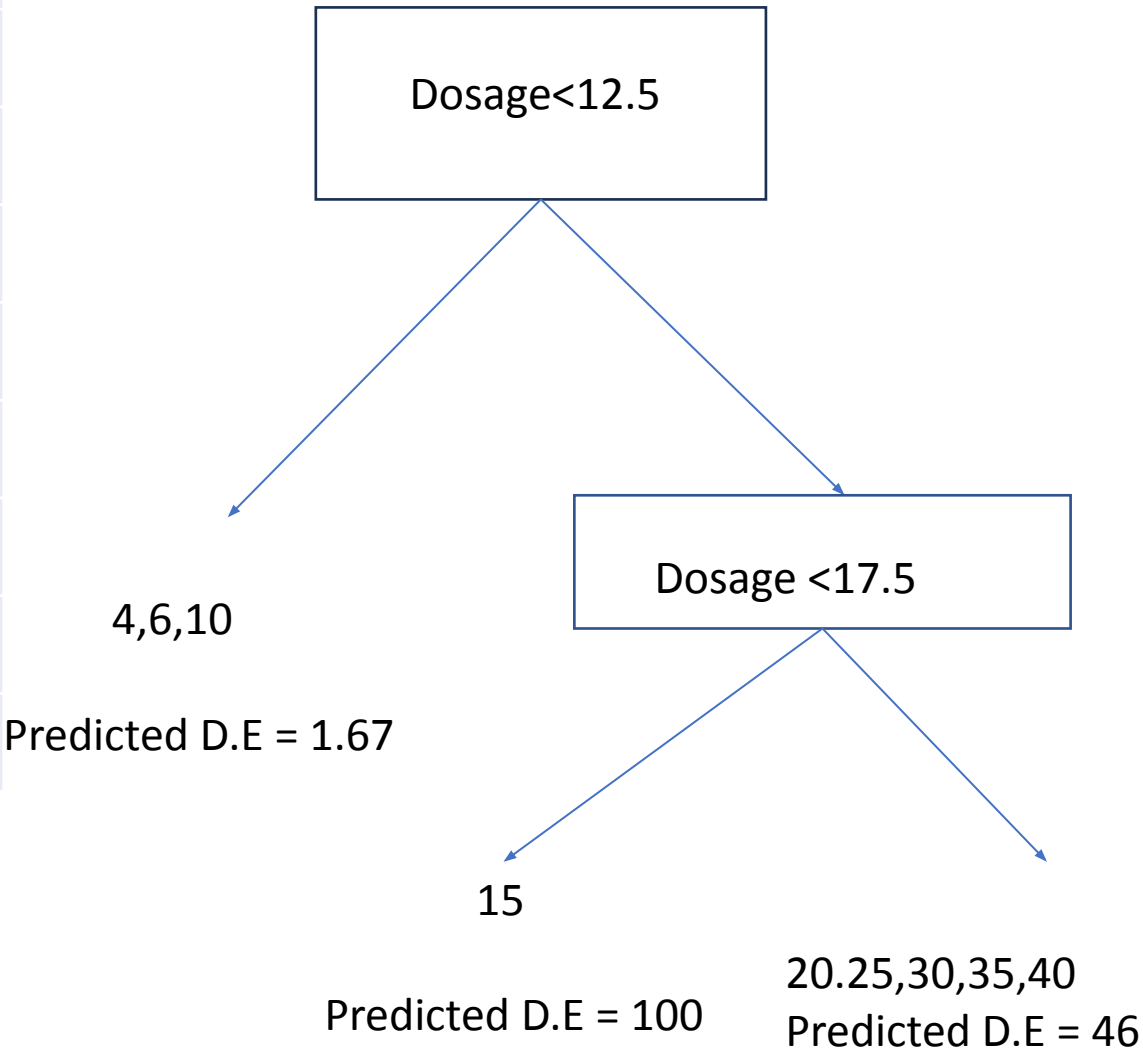
Dosage	Gender	Drug Effectiveness
4	M	0
6	M	2
10	F	3
15	F	100
20	M	100
25	M	100
30	F	20
35	F	10
40	M	0

Threshold	SSR
5	16485
8	14670
12.5	12355
17.5	17177
22.5	17883
27.5	14709
32.5	15374
37.5	16485



Dosage	Gender	Drug Effectiveness
4	M	0
6	M	2
10	F	3
15	F	100
20	M	100
25	M	100
30	F	20
35	F	10
40	M	0

Threshold	SSR
17.5	9920
22.5	6275
27.5	
32.5	
37.5	

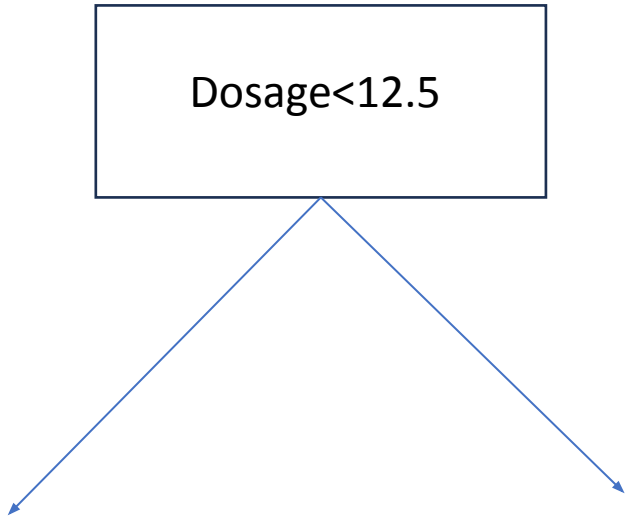


SSR (Dosage<17.5) = (100-100)^2+
(100-46)^2+(100-46)^2+(20-46)^2+(10-46)^2
+(0-45)^2 = 9920

SSR(Dosage<22.5) = (100-100)^2+(100-100)^2
+(100-32.5)^2+(20-32.5)^2+(10-32.5)^2+(0-32.5)^2 = 6275

Dosage	Gender	Drug Effectiveness
4	M	0
6	M	2
10	F	3
15	F	100
20	M	100
25	M	100
30	F	20
35	F	10
40	M	0

Threshold	SSR
17.5	9920
22.5	6275
27.5	200
32.5	7425
37.5	13076



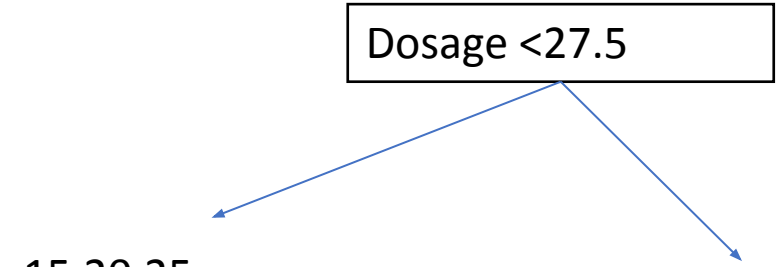
4,6,10

Predicted D.E = 1.67

15,20,25,30,35,40
Predicted D.E = 55

SSR(Dosage<27.5) =
(100-100)^2+(100-100)^2+(100-100)^2+(20-10)^2+(10-10)^2+(0-10)^2
= 200

SSR(32.5) = (100-105)^2+(100-105)^2+(100-105)^2+(20-105)^2
+(10-5)^2+(0-10)^2 = 7425
SSR(37.5) = (100-66)^2+(100-66)^2+(100-66)^2+(20-66)^2+

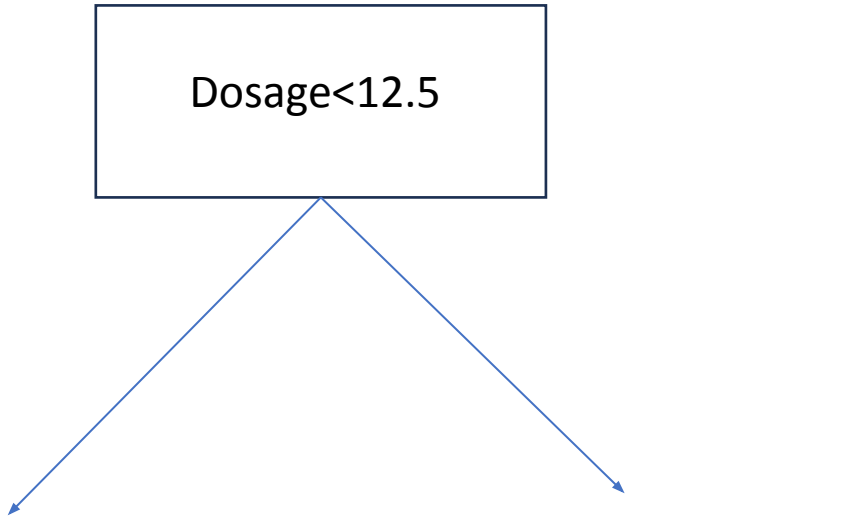


15,20,25
Predicted D.E = 100

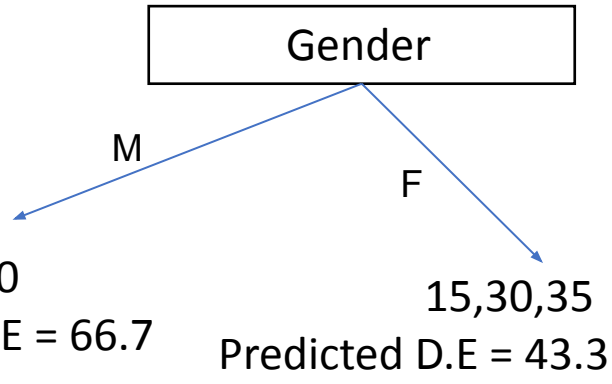
30,35,40
Predicted D.E = 10

Dosage	Gender	Drug Effectiveness
4	M	0
6	M	2
10	F	3
15	F	100
20	M	100
25	M	100
30	F	20
35	F	10
40	M	0

Threshold	SSR
17.5	9920
22.5	6275
27.5	200
32.5	7425
37.5	13076



SSR(Gender) =
(100-66.7)^2+(100-66.7)^2+(0-66.7)^2+(100-43.3)^2+(20-43.3)^2+(10-43.3)^2 = 6666.67+3214.89+542.89+1108.89
= 10990.45



Hence among Gender and Dosage < 27.5, Dosage <27.5 is the winner

Dosage	Gender	Drug Effectiveness
4	M	0
6	M	2
10	F	3
15	F	100
20	M	100
25	M	100
30	F	20
35	F	10
40	M	0

